

This section establishes the technical foundation through prerequisites validation and workspace creation. Participants verify Kiro CLI installation, authentication, and required tools (Python 3.7+, JSON validation) using validation commands. The setup creates an organized workspace with isolated virtual environments for development (src/to-do-app/venv) and testing (tests/venv), plus shared collaboration directories for seamless file-based communication between agents. This ensures all participants start with identical, properly configured environments for the multi-agent demonstration.

Prerequisites Check & Validation

Required Software:

- Kiro CLI installed and authenticated
- Basic understanding of JSON configuration
- Terminal access (bash/zsh)
- Text editor (VS Code recommended for JSON validation)
- Python 3.7+ with pip (for QA testing capabilities)

Validation Script - Run These Commands:

1. Check Kiro CLI installation

```
1 # Expected: Kiro CLI version x.x.x
2 kiro-cli --version
```



2. Verify authentication status

```
1 # Expected: Logged in with IAM Identity Center (https://your-org.awsapps.com/start)
2 # Profile: your-profile-name
3 # arn:aws:codewhisperer:us-east-1:xxxxxxxxxxxx:profile/XXXXXXXXXXXXX
4 kiro-cli whoami
```



3. Test basic functionality

```
1 # Expected: Help text showing chat command options
2 kiro-cli chat --help
```



4. Test JSON syntax knowledge

```
1 # Expected: Formatted JSON output (or use `jq` if available)
2 echo '{"test": "valid"}' | python -m json.tool
```



If any validation fails:

- Kiro CLI not found: [Install from AWS documentation](#)
- Authentication issues: Run `kiro-cli login --use-device-flow` and follow the prompts
- JSON validation: Install VS Code with JSON extension
- Python/pip issues: Install Python 3.7+ and ensure pip is available

Workshop Workspace Setup

Create the project structure

1. Create working directory

```
1 mkdir /Workshop/simple-todo-app && cd /Workshop/simple-todo-app
```



2. Create shared collaboration space

```
1 mkdir -p shared
2 mkdir -p contexts           # Context files for agent resources
3 mkdir -p .kiro/agents       # Agent JSON configurations
4 mkdir -p src/todo-app       # Application with its own venv
5 mkdir -p tests              # Testing with its own venv
```



3. Set up virtual environment for the application

```
1 cd src/todo-app \
2 && python3 -m venv venv \
3 && source venv/bin/activate \
4 && pip3 install flask \
5 && deactivate \
6 && cd ../..
```



4. Set up virtual environment for testing

```
1 cd tests \
2 && python3 -m venv venv \
3 && source venv/bin/activate \
4 && pip3 install pytest requests \
5 && deactivate \
6 && cd ..
```



Initialize shared files

1. Create simple collaboration files

```
1 touch shared/requirements.md # What we need to build
2 touch shared/status.md       # Development progress
3 touch shared/bugs.md         # Issues found during testing
4 touch shared/test-report.md  # QA test results
5 touch shared/uat-feedback.md # User acceptance feedback
```



Checkpoint: Validate Your Setup

1. Verify directory structure

```
1 # Expected: agents/ directory
2 ls -la .kiro/
```



2. Check shared files

```
1 # Expected: 5 .md files created (requirements, status, bugs, test-report, uat-feedback)
2 ls -la shared/
```



3. Check contexts directory for agent resources

```
1 # Expected: Empty directory (will be populated with context files)
2 ls -la contexts/
```



4. Check virtual environments are set up

```
1 # Expected: Virtual environment directories with bin/, lib/, etc.
2 ls -la src/todo-app/venv/
3 ls -la tests/venv/
```



5. Confirm you're in the right directory

```
1 # Expected: /path/to/simple-todo-app
2 pwd
```

